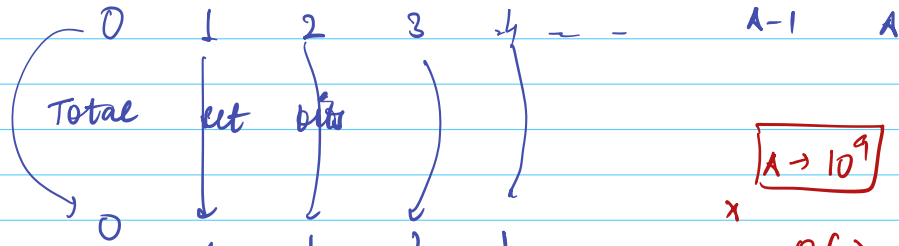


(A)



0th

Bit Position	Bit Value	Count
0	0	0
1	0	1
2	1	2
3	1	3
4	0	4
5	1	5
6	0	6
7	1	7
8	0	8
9	1	9
10	0	10
11	1	11
12	0	12
13	1	13
14	0	14
15	1	15
16	0	16

16th bit

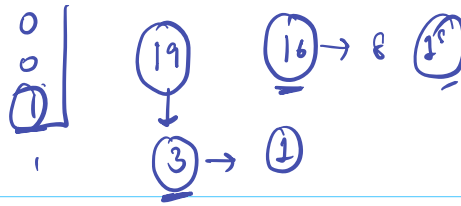
How many times 0th bit will be set till A.

Total number
N = A + 1

$$N/2 \checkmark$$

$$\frac{2^0}{2}$$

$$N = 17$$

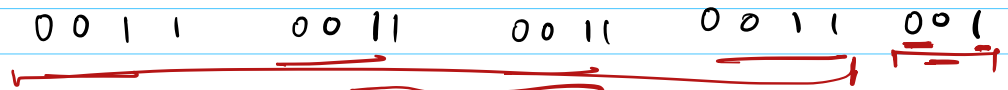


$2^1 \rightarrow$

Set size = 4

$$N/4 \times 2 + ?$$

$$\text{left over numbers} = N - \underbrace{\left(N/4\right) \times 4}_{N/4}$$



$$19 \div 4 = 3$$

✓ $\left(\frac{19}{4} \times 2\right)$

number of set $\times 2$

$$4 \rightarrow 2$$

$$3 - 2 = 1$$

$$\underline{3} \geq \underline{2}$$

$$\downarrow$$

$$\underline{3-2}$$

2^2

0000 1111 0000 1111 000

$$19/8 * 4 = 8$$

$$19 \div 8 = 3$$

8

4

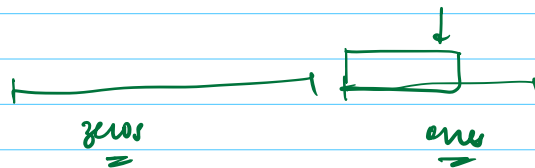
1^8
2

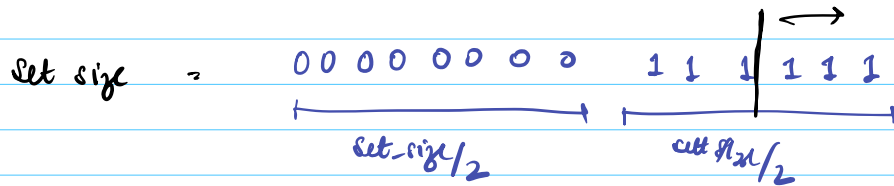
$$\left\lfloor \frac{N}{\text{set size}} * \left(\frac{\text{set size}}{2} \right) \right\rfloor$$

$$\underline{N \% \text{ set size}} < \frac{\text{set size}}{2} \rightarrow 0$$

else

$$\left(N \% \text{ set size} - \frac{\text{set size}}{2} \right)$$





19

0	0	0	0	0
0	0	0	0	1
0	0	0	1	0
0	0	0	1	1
0	0	1	0	0
0	0	1	0	1
0	0	1	1	0
0	0	1	1	1
0	1	0	0	0
0	1	0	0	1
0	1	0	1	0
0	1	0	1	1
0	1	1	0	0
0	1	1	0	1
0	1	1	1	0
0	1	1	1	1
1	0	0	0	0
1	0	0	0	1
1	0	0	1	0
1	0	0	1	1

set size = 2

$$20/2 \rightarrow 10 \rightarrow 10 \text{ is}$$

(21) set size = 4 → 2

$$20/4 = 5 \rightarrow *2 = (10)$$

$$20 \div 4 \rightarrow$$

(22)

$$20/8 \rightarrow 2 \rightarrow 2 * 4 = (8)$$

$$20 \div 8 = (4) \times$$

(23) (16)

$$20/16 \rightarrow 1 \rightarrow 8 \rightarrow (8)$$

$$20 \div 16 \rightarrow 4 \checkmark \rightarrow 0$$

$$20/32 \rightarrow 0$$

$$20 \div 32 \rightarrow (20) - 16 = (4) \checkmark$$

16 ✓

Smallest XOR!

(A)

(B)

$$\min(A \oplus x)$$

number of set bits = B ✓

$A \oplus x \rightarrow$
 $\downarrow$
 (A)

? ?
 $\downarrow \downarrow$
 (1) 0 1 0 1 1 1 0

$x \rightarrow 1 \quad 1$

4 bits

$x \rightarrow 1011$

$x = B = 001000$

19 $\rightarrow$ 10011

4 bits

x 1011

$19 \oplus 16$

$19 \oplus 2$

B == no. of bits(A)

Ans = A

Observation Always try to set MSB on priority!

$B <$ no. of set bits A
 $\downarrow$

$B >$ no. of bits A

(A) $\rightarrow$ then start

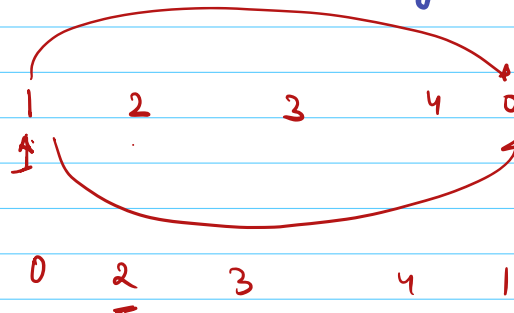
✓ $\downarrow$ Try to turn on MSB in x

✓ turning on bits from MSB to LSB

0 0 0 0 0

Minimum swap 2

Aim: To sort the array



$O(N)$
 $\downarrow$
Sort the array Hashmap

$0 \rightarrow N-1$!

$0 \rightarrow N-1$,

while ($i < n$) :

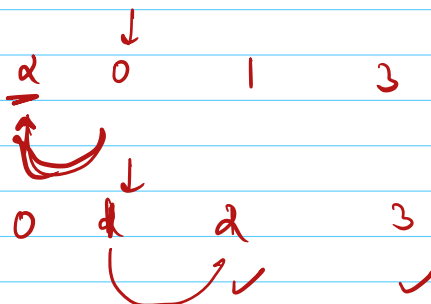
if ($arr[i] \neq i$)

swap (i , $arr[i]$)

$O(N)$

$O(N)$

cnt++
cnt++



$arr[i], i$

$f(x, y) \rightarrow$ no. of different corresponding bits in x, y

$2 \rightarrow (0, 1, 0)$
 $7 \rightarrow (1, 1, 1)$ $(2) \Rightarrow arr$

$\sum f(arr[i], arr[j])$ (i, j)

	1	3	5
	001	011	101
1	0	1	1
3	1	0	2
5	1	2	0

(N^2)
 $N \rightarrow 10^5$

(8)

2^0 1 1 1

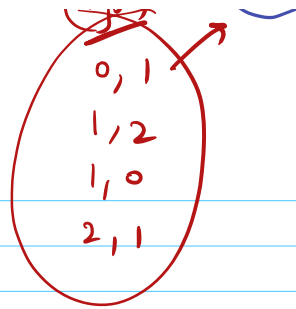
1
1
1

(x)

$[1, 1, 1] \rightarrow$ no pairs with different bit

2^1

(i, j) $[0, 1, 0]$ $[4 \text{ pairs}]$

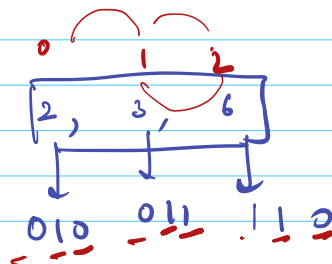


$2^{\text{count_set_bits}}$
 $\times \text{count_unset_bits}$

2^2 [0 0 1]

$2^{\text{count_set_bits}} \times \text{count_unset_bits} \rightarrow 4 \quad 8$

i, j ~~$i < j$~~



$2^0 \rightarrow 4 \text{ pairs}$

$2^1 \rightarrow 0 \text{ pairs}$

$2^2 \rightarrow 4 \text{ pairs}$

78

for ($i = 0$; $i < 32$; $i++$)

int cnt_set_bits = 0

for ($j = 0 \rightarrow n$)

if ($A[j] \& (1 < i)$)

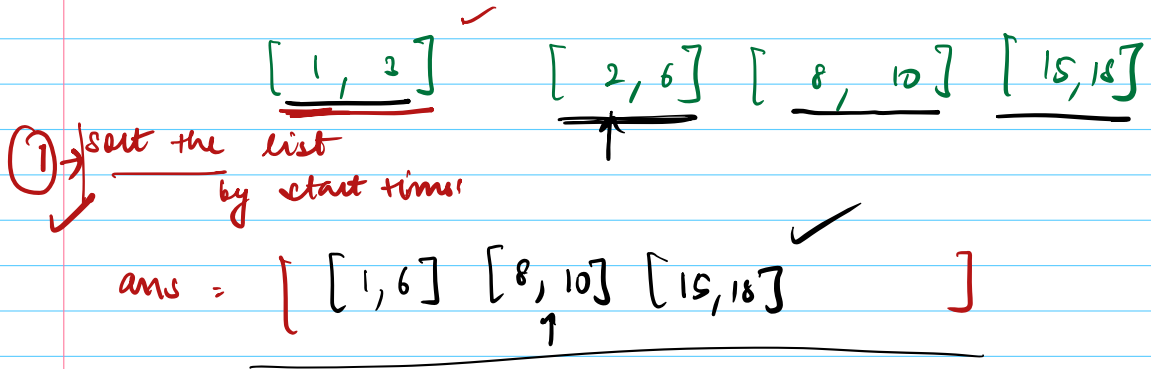
cnt_set_bits ++

ans += $2^{\text{cnt_set_bits}} \times (n - \text{cnt_set_bits})$

i, j
 $0, 1$

10

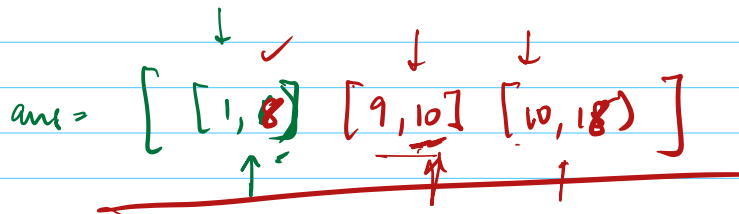
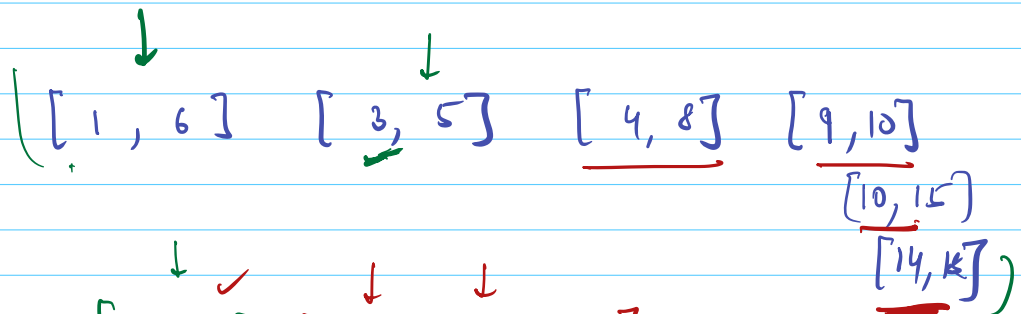
Merge Overlapping Intervals.



① check if overlapping with last inserted interval.

$$\max(\text{end}_i, \text{end}_{i+1})$$

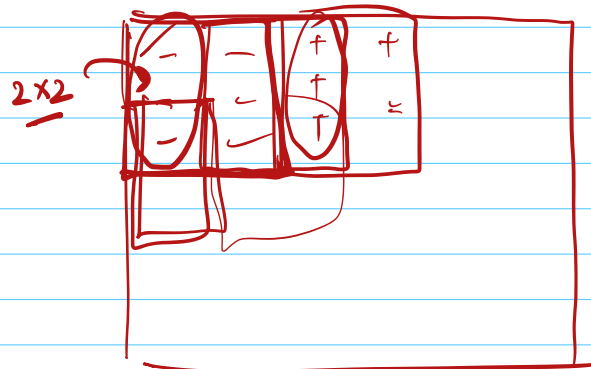
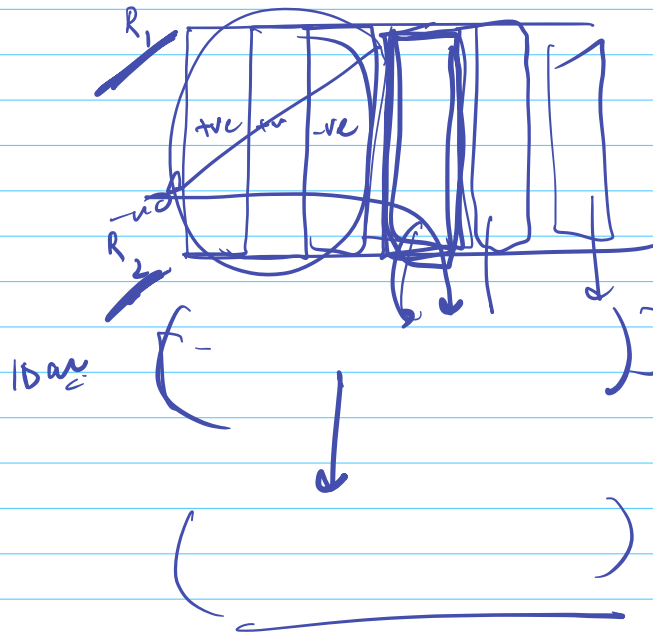
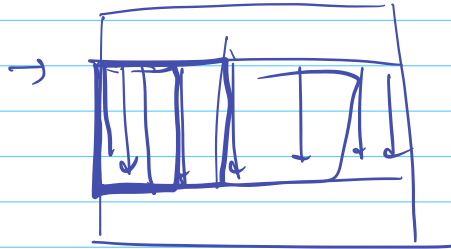
list < intervals >



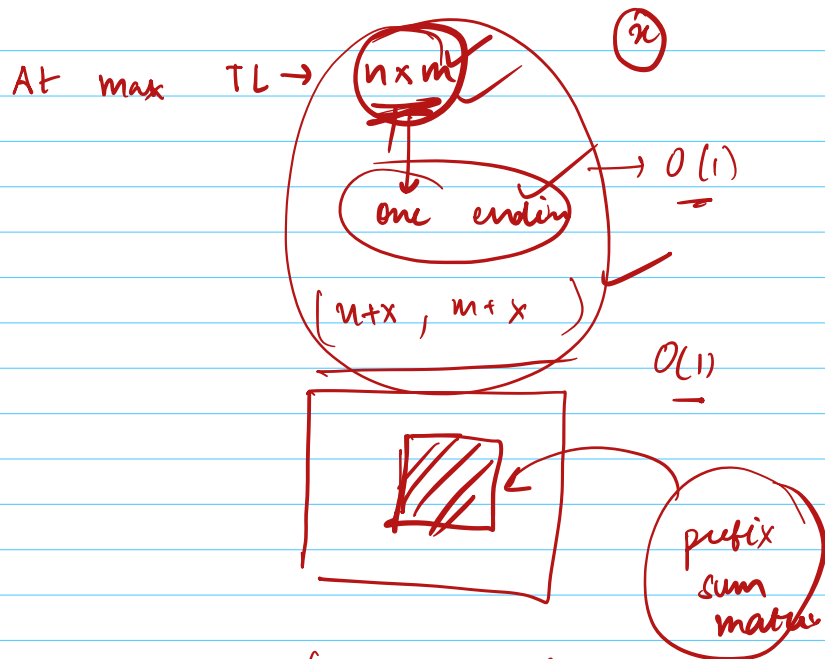
$$\text{start}_i \leq \text{end}_{\text{last}}$$



5 min



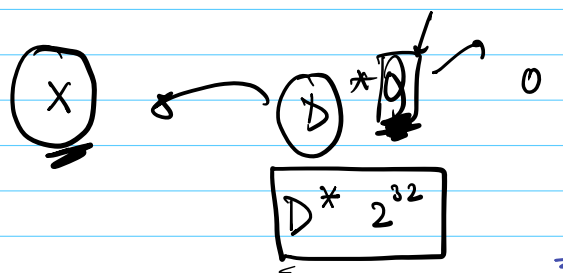
$O(N * M)$



for $(i \rightarrow 0 \rightarrow n)$
 $(j \rightarrow 0 \rightarrow n)$

$(i, j) \rightarrow (i+x, j+x)$

ans -



$$\begin{array}{r} 150 \\ -102 \\ \hline 48 \\ -28 \\ \hline 20 \\ 14 \rightarrow \end{array}$$

$$7 \times 16 = 112$$

$$7 \times 2^5 = 224$$

$$7 \times 2^3 = 56$$

$$7 \times 2^2 = 28$$

$$2^4 + 2^2 + 2^1 = 22$$

$$22 \times 22 = 484$$

$$7 \times 22 = 154$$

$$\frac{12}{6} = 3$$

$$7 \times 16 =$$

$$1 \times \frac{16}{7} = \frac{112}{7}$$

$$7 \times 2 = 14$$

$$\frac{32}{3}$$

$$x \neq A$$

$$A \wedge x = A + x$$

$$y \geq A$$

$$A \wedge y = A + y$$

$$5 = \begin{array}{r} 101 \\ 010 \\ \hline \end{array}$$

$$\begin{array}{r} 101 \\ 101 \\ \hline \end{array}$$

$x \oplus r = \text{sum}$

$$\textcircled{c} \begin{array}{r} 101 \\ 1000 \\ \hline \end{array}$$

$$\begin{array}{r} 11 \\ 01 \\ \hline \end{array}$$

$$x \rightarrow \boxed{\text{MSB} \rightarrow 0}$$

$10 =$

Hint when?

$$\wedge = +$$

$$\checkmark \boxed{1}$$

find no.s

$$\underline{A \wedge p}$$

